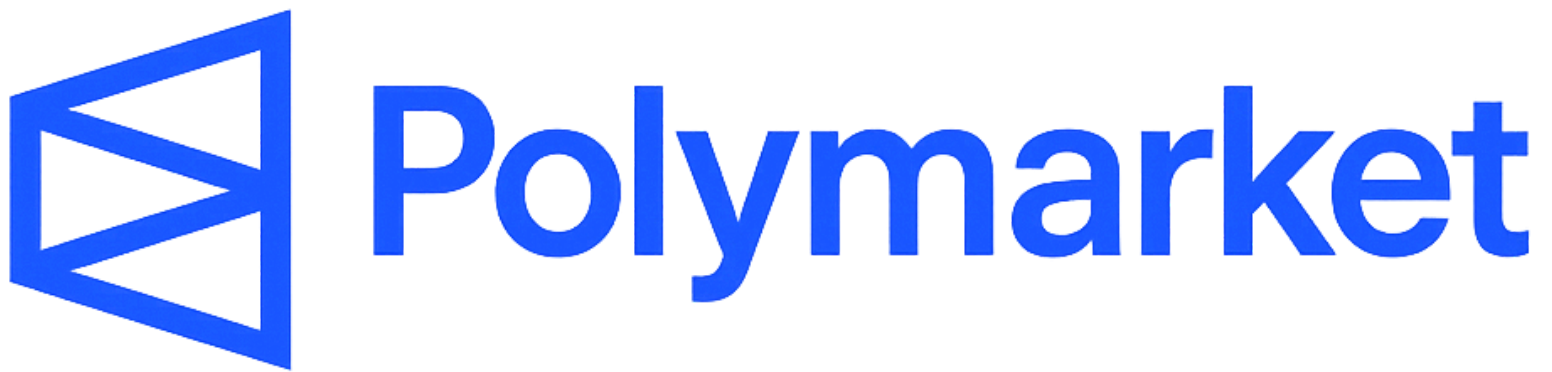


Did traders who placed bets early in the
**"Will there be a US Government shutdown
before 2025?"** market on Polymarket
achieve the same ROI as late traders?



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ABSTRACT

Polymarket is a prediction market platform with thousands of markets (real-world events that you can bet on). Each market has two outcomes: Yes and No. Traders can buy and sell shares of each outcome—which will move that outcome's price—prior to the "resolution" of an event. At resolution, shares of the correct outcome become worth \$1, and shares of the incorrect outcome become worth \$0. This observational study investigated whether "early" traders in a specific Polymarket prediction market achieved different Returns on Investment (ROI) than "late" traders. The "Will there be a US Government shutdown before 2025?" market—and the 5,080 unique wallets (traders) who participated in this market—were used for this case study. This market ran from September 3, 2024 to December 31, 2024, and each wallet was classified as either early or late based on whether their first trade occurred before or after December 17, 2024 at 7:00 P.M. EST. This timestamp marks a notable shift in public sentiment towards the outcome of this market. Two independent simple random samples of 250 wallets were obtained for each group (meaning 500 wallets were examined in total) through Dune, and each wallets' ROI was calculated after rebuilding their trade history through Polymarket's API. Then, a two-sample t-test was used to see whether or not there was a difference in the average ROI between early and late traders. Also, a 95% two-sample t-interval for $\mu_{\text{early}} - \mu_{\text{late}}$ was constructed to estimate the magnitude of any difference in mean ROI between early and late traders.

Hypotheses

Two-Sample T-Test:

$$\alpha = 0.05$$

μ_{early} = the true mean ROI for all traders who completed their first trade on the "Will there be a US Government shutdown before 2025?" market on or before 12/17/24 at 7:00 P.M. EST.

μ_{late} = the true mean ROI for all traders who completed their first trade on the "Will there be a US Government shutdown before 2025?" market after 12/17/24 at 7:00 P.M. EST.

Null Hypothesis (H_0): There is no difference between the average ROI of early and late traders ($\mu_{\text{early}} = \mu_{\text{late}}$)

Alternative Hypothesis (H_a): There is a difference between the average ROI of early and late traders ($\mu_{\text{early}} \neq \mu_{\text{late}}$)

Two-Sample T-Interval:

Confidence Level: 95%

Parameter of Interest: $\mu_{\text{early}} - \mu_{\text{late}}$; the true difference in mean ROI between early and late traders.

For clarity, μ_{early} and μ_{late} are defined above (found under *Two-Sample T-Test*)

Goal: To estimate the true difference in the mean ROI between early traders and late traders in the "Will there be a US Government shutdown before 2025?" Polymarket prediction market at a 95% confidence level.

Study Design: How

Step #1: Dune (a crypto analytics platform that provides blockchain datasets) was used to gather the wallet address of each trader who traded on the market named above, and a corresponding timestamp for when they completed their first trade (on this specific market).

Step #2: The traders were then divided into two groups based on their first trade's timestamp relative to *12/17/24 at 7:00 P.M. EST*. If their first trade was on or before that time, then they were classified as early. Otherwise, they were classified as late.

Step #3: A unique randomly generated number was assigned to every wallet within both the "early" and "late" groups. Using a random number generator again, 250 random numbers were obtained for each group. The data containing all of the wallets, their assigned group, and their randomly assigned number was filtered to only include the wallets whose random numbers corresponded with the randomly generated ones. After filtering, there were 250 randomly sampled wallets in the "early" group and 250 randomly sampled wallets in the "late" group.

Step #4: Polymarket's API was used to reconstruct the full trade history (within this specific market) for each user. Specifically, for each trade, the following information was obtained to calculate the ROI of the wallet: the outcome (Yes/No), the side (Buy/Sell), the number of shares, and the cost/price of 1 share.

Step #5: After obtaining each wallet's full trade history (for the market), the ROI (for each wallet) was calculated with the following formula:

$$ROI = \frac{(S + L) - B}{B} \cdot 100$$

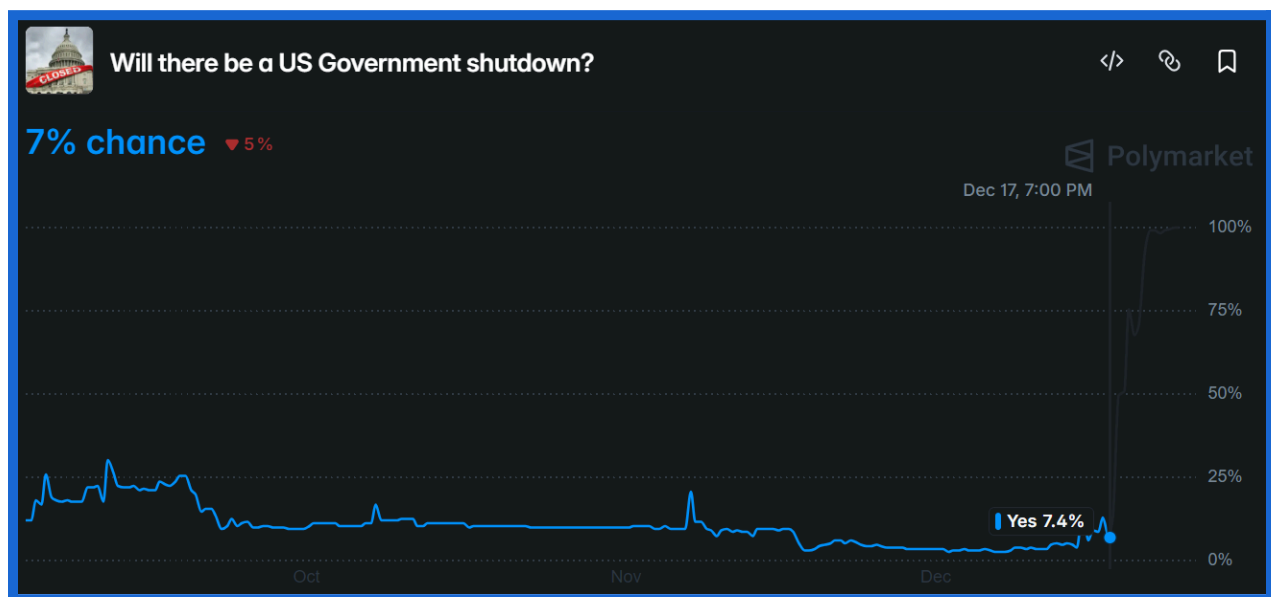
Where:

- S = Sum of every (sale price * # of shares sold) when the wallet sold shares for either outcome
 - Both outcomes (Yes/No) are calculated separately and then summed together
- L = Left over value of shares at resolution
 - When the market resolves to either Yes or No, shares that match the outcome of the market become worth \$1, and shares that do not match the outcome become worth \$0
- B = Sum of every (buy price * # of shares bought) when the wallet bought shares for either outcome
 - Both outcomes (Yes/No) are calculated separately and then summed together

Study Design: Why

Why this market?: This market was picked for three reasons. The first is that I wanted a market with a relatively small volume (this one had \$54M) as there'd be less trades to process (I wasn't sure if I'd get rate limited by the API). The second reason is that this market was not tied to an "event". An event on Polymarket has many markets tied to it (e.g. event: Who will be the next president? // markets for that event: [candidate #1 with Yes/No outcome], [candidate #2 with Yes/No outcome], etc). The final reason was that the market was in the politics category. I think that category is better to analyze than something like sports, as there is theoretically more certainty in the outcomes (traders aren't as likely to be gambling), especially prior to resolution.

Why the timestamp?: This time was hand-picked as it is right before the outcome rapidly switched from No to Yes (right before resolving to Yes), indicating new information was released to the public that strongly shifted traders' opinions on the resolution. I initially considered a share-volume based split (e.g. if a market has 200,000 total times that Yes/No shares were transferred between buyers and sellers, then split on the 100,000 mark). This was unfortunately too difficult for me to figure out how to do this on time *and* it had a big limitation, which is addressed under *Limitations & Potential Improvements* in the *Conclusion* section.



Why Dune?: Polymarket handles thousands—if not hundreds of thousands—of trades among its many markets each second. Unfortunately, they do not do a good job of providing this raw data in an easily accessible format, despite the fact that it's all public. Dune bridges the gap between Polymarket's blockchain activity and meaningful data by indexing on-chain transactions and making them accessible through SQL queries. This allows researchers to filter, aggregate, and analyze trade data at scale without needing to parse through raw blockchain data directly

(through other means). Prior to this project, I was unfamiliar with SQL, which is why I didn't bother to reconstruct wallet trade histories (besides its first trade on the market) through Dune.

Why Polymarket's API?: Following up on the information above, Polymarket's API was more intuitive for reconstructing trade histories (and calculating ROI from that) *once I had the wallet addresses*, since I am familiar with Python.

Why calculate ROI *that way*?: In a traditional investment, ROI is straightforward—you spend money, sell your position, and compare what you got back to what you put in. Prediction markets are slightly more complex because a trader's total return comes from two distinct sources: shares they sold before the market resolved (S), and leftover shares (shares still held) at resolution (L). Ignoring either source would produce an incomplete picture of a trader's performance. For example, a wallet that never sold a single share but held the winning token would *appear* to have zero proceeds *despite* earning a return. By combining S and L and measuring both against total spending (B), this formula captures the full financial outcome of a wallet's activity within this market regardless of their trading strategy.

Raw Data

As the data has 500 rows, it was submitted separately. A spreadsheet of the data can be found [HERE](#). Alternatively, you can download the .csv file [HERE](#) (assuming I don't private the repository in the future). That repository also has all the code that was used for this project for data collection.

The image below is a sample of what the dataset looks like. Group 'A' was the early group, and group 'B' was the late group. The rightmost column was used for the data as it shows the ROI as a percentage (as I forgot to multiply it by 100 within my code. This issue was solved by doing it in sheets instead). Note the row numbers as well—the 500 total wallets are found in rows numbered from 2 to 501.

1	address	group	#	roi	+	ACTUAL ROI (as a %)
2	0x02584443ceee8c2a77a1bf54572d8376d20654	A		-1		-100
3	0x065eedbf3509a041f104aeced3966da1b85dcf2	A		-1		-100
4	0x07211f011080a4bf3f5282cd64a71afc87e54a8	A		-1		-100
5	0x07b34a692c3689b0f7e9265eb6658b3cad0534	A		-1		-100
6	0x1347d5a2a619363efda8ff766e6297fbe8099f2c	A		-1		-100
7	0x1be8d430ff58de57fbc99989e70908168d8d0d1	A		-1		-100

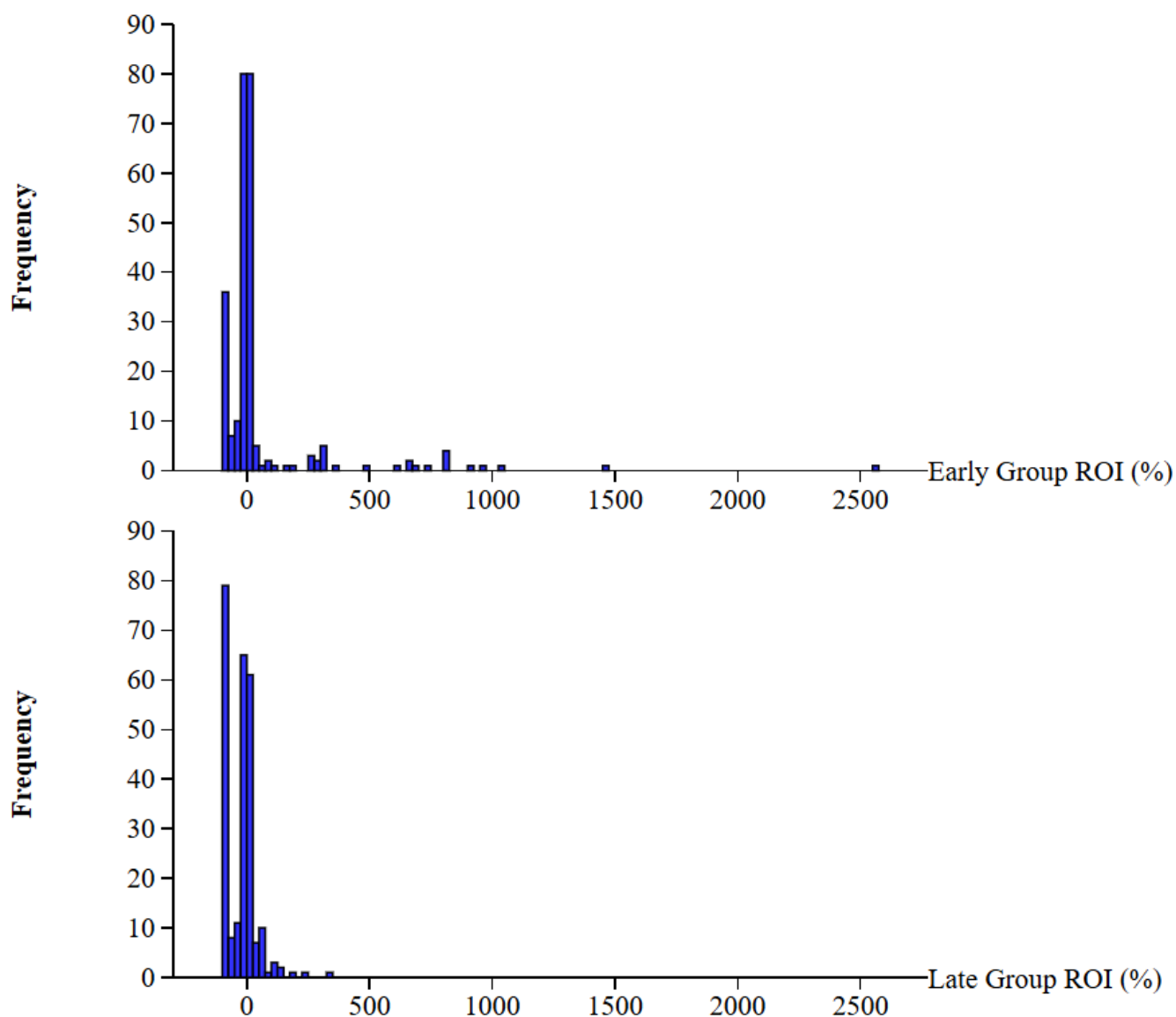
AND

250	0xcf015afd93d661fdde782d59e1bf8c87e80173e1	A		14.58544556		1458.54
251	0xfd13172de98a7dff6fb054107765470c30e1e6f	A		25.50306024		2550.31
252	0x04dcbd9b805b220a9b2d27e4d9e649ed07902c	B		-1		-100
253	0x09e6d660304b9590acd4910f17af31f5ed8a24c	B		-1		-100
254	0x101f4a6685af466b880333af08f865f3a285507c	B		-1		-100
255	0x113e974768d5731468dc7b1c62e58dd854b72c	B		-1		-100
256	0x1446fddafd68b9b83717e91d5ab35da6db247fc	B		-1		-100

Data Analysis

Data Summary:

Note: Everything in this section was obtained via stapplet.com.



Group Name	n	mean	SD	min	Q ₁	med	Q ₃	max
1: Early Group ROI (%)	250	55.227	264.269	-100	-14.29	-0.1	4.16	2550.31
2: Late Group ROI (%)	250	-25.215	63.183	-100	-100	-0.135	0.5	345.68

Data Commentary:**Histograms:**

Both histograms look relatively similar, although it is noticeable that the late group has many more wallets with an ROI of -100% than the early group. The late group has roughly 80 of those types of wallets, while the early group has less than 40 of those types of wallets. Also, the early group appears to have many more wallets that achieved ROIs that are greater than 500%. This could be indicative that the average ROI for the early group will be much higher than the late group, especially with the possible outliers (with large ROIs) that might pull the average upwards.

5 Number Summary:

Early Group: The early group has a mean ROI of 55.227%, which at first glance looks like most early traders were very profitable. However, this conclusion becomes questionable when accounting for potential outliers on the high end (such as the max—a trader with an ROI of 2550.31%). The median of -0.1% reveals that the distribution is heavily right-skewed, which explains why the mean is pulled so far upward. Additionally, the Q1 of -14.29% tells us that the bottom 25% of early traders exited the market at a loss. The Q3 of 4.16% tells us that 75% of early traders had an ROI at or below 4.16%—a very modest return for traders at such a high percentile. This gives the early group an IQR of 18.45, meaning the middle 50% of early traders were clustered in a narrow 18.45 percentage point range, despite the group's massive standard deviation of 264.269%. This contrast highlights just how severely a small number of potential outliers are inflating the standard deviation. Overall, this data tells us that a small number of extremely profitable wallets are pulling the mean ROI far above what most early traders actually experienced. In reality, the majority of early traders lost money or profited a very small amount.

Late Group: The late group has a mean ROI of -25.215%, which at first glance looks like most late traders lost a good proportion of their money. However, this conclusion becomes questionable when accounting for the median of -0.135%, which indicates that most of the traders in the late group did not lose nearly as much money as the average implied. After looking at the Q1 (-100%), everything becomes clearer. At least 25% of the late traders lost everything, which explains why the distribution is extremely left skewed, and why the mean is so much less than the median. Additionally, the IQR of 100.5 percentage points is deceptively large—it is almost entirely the result of Q1 being so low, meaning the spread in ROIs of the middle 50% is driven by many heavy losers on the bottom end rather than big winners on the top (since the Q3 is a very small return of 0.5%). The max ROI of 345.68% and the standard deviation of 63.183% confirm that some (even if it was a very select few) late traders did in fact profit, and that total-loss wallets dragged the distribution downward. Overall, the mean ROI of -25.215% is misleading—in reality, the typical (median) late trader barely lost money (not 1/4 of their investment), but a substantial subset of traders who lost everything are pulling the average far below what most late traders actually experienced.

Early and Late Groups Compared: Despite having very different means—55.227% for the early group versus -25.215% for the late group—both groups tell a surprisingly similar story when looking at their medians. The typical (median) early trader had an ROI of -0.1%, while the typical (median) late trader had an ROI of -0.135%, meaning the middle trader in both groups both lost a very small (and similar) amount of money relative to their investment. This suggests that the dramatic difference in means is not reflective of a dramatic difference in the typical trading experience, but rather a product of how each group's wallets pull their respective averages in opposite directions. The key distinction between the two groups lies in the direction

of their skew. The early group is heavily right-skewed, with a max of 2550.31% and a standard deviation of 264.269%—a small number of enormously profitable wallets inflate the mean far above what most early traders experienced. The late group, on the other hand, is heavily left-skewed, with 25% of the wallets having an ROI -100% dragging the mean well below what the typical (median) late trader experienced. In other words, the early group's mean is distorted by a few massive winners, while the late group's mean is distorted by a large number of total losers. The IQR comparison further illustrates this. The early group's IQR of 18.45 percentage points is notably narrow, meaning the middle 50% of early traders had tightly clustered outcomes despite the group's enormous standard deviation. The late group's IQR of 100.5 percentage points appears much wider, but as previously discussed, this is almost entirely driven by Q1 being -100% rather than by any meaningful profitable traders at the top (since Q3 was only 0.5%). In both cases, the IQR is being distorted, but it's in opposite directions.

Statistical Tests: Two-Sample T-Test

Why a Two-Sample T-Test?:

A two-sample t-test is appropriate because we are comparing the means of a quantitative variable (ROI) across two independent groups (early and late traders). Since the true population standard deviations are unknown, we estimate them using sample standard deviations and use the t-distribution to test a claim (whether $\mu_{\text{early}} = \mu_{\text{late}}$).

Conditions:

Random: The *Study Design: How* section details exactly how each random sample was selected; two Simple Random Samples, each with 250 wallets, were obtained.

Normal: Both groups have $n = 250$ wallets each, which are both ≥ 30 . Based on this information, the Central Limit Theorem guarantees that the sampling distribution is approximately normal.

10%: This condition is not fully met, and this is addressed in the limitations section. While 500 total wallets is $\leq 10\%$ of the 5,080 unique wallets that traded on the specified market, the sampling process pulled from each individual group. This means that 4,578 late wallets were sampled from, and 502 early wallets were sampled from without replacement. It's also important to note that it is assumed that each wallet is independent from each other (no user has more than one wallet). However, we will proceed with the test with caution.

- Early group: $250 \leq 0.10(502)$ is *NOT* a true statement as $250 \geq 50.2$
- Late group: $250 \leq 0.10(4578)$ is a true statement as $250 \leq 457.8$

Calculations:

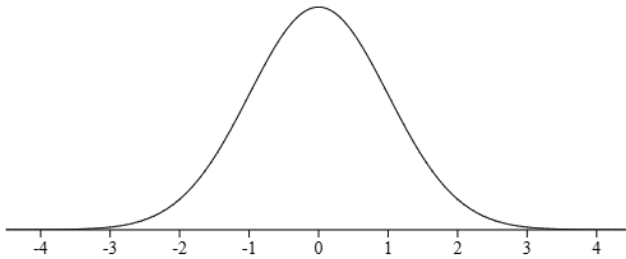
$$t = \frac{(\bar{x}_{early} - \bar{x}_{late}) - 0}{\sqrt{\frac{s_{early}^2}{n_{early}} + \frac{s_{late}^2}{n_{late}}}}$$

$$t = \frac{55.227 - (-25.215)}{\sqrt{\frac{264.269^2}{250} + \frac{63.183^2}{250}}}$$

$$t = 4.681$$

$$df = 250 - 1 = 249$$

$$p = 2 \cdot tcdf(4.681, 1E99, 249)$$



$$p = 2 \cdot 0 = 0$$

Note: d.f. used conservative approach.

Mathematical Conclusion:

Since $p(0.0) < \alpha(0.05)$, we reject H_0 . There is convincing evidence to conclude that the true mean ROI for "early" traders is different from the true mean ROI for "late" traders in the "Will there be a US Government shutdown before 2025?" prediction market on Polymarket. However, since the 10% condition wasn't met, there is more to be concluded about the results of this test. This is addressed in the *Conclusion* section.

Statistical Tests: Two-Sample T-IntervalWhy a Two-Sample T-Interval?:

A two-sample t-interval is appropriate because we are estimating the magnitude of the difference in means of a quantitative variable (ROI) between two independent groups (early and late traders). Since the true population standard deviations are unknown, we use a t-interval (rather than a z-interval) to construct a 95% confidence interval for the parameter $\mu_{\text{early}} - \mu_{\text{late}}$.

Conditions:

For the sake of simplicity, the conditions will not be restated here again because the conditions for the interval (and their checks) are identical to the conditions for the two-sample t-test above. This includes the fact that the 10% condition was *not* met, but again, we will proceed with caution when creating the interval.

Calculations:

$$t^* = \text{inv}T(0.975, 249) = 1.9695$$

$$CI = (\bar{x}_{\text{early}} - \bar{x}_{\text{late}}) \pm t^* \sqrt{\frac{s_{\text{early}}^2}{n_{\text{early}}} + \frac{s_{\text{late}}^2}{n_{\text{late}}}}$$

$$CI = (55.227 + 25.215) \pm 1.9695 \sqrt{\frac{264.269^2}{250} + \frac{63.183^2}{250}}$$

$$CI = (46.5963, 114.2877)$$

Note: d.f. used conservative approach.

Mathematical Conclusion:

We are 95% confident that the true difference in the mean ROI between early traders and late traders ($\mu_{\text{early}} - \mu_{\text{late}}$) in the "Will there be a US Government shutdown before 2025?" prediction market on Polymarket lies between 46.5963% and 114.2877%. As 0% is far outside the interval, we have strong evidence to conclude that the average ROI for early traders is significantly higher than the average ROI of late traders. However, since the 10% condition wasn't met, there is more to be concluded about the results of this test. This is addressed in the *Conclusion* section.

Conclusion

Implications of the Statistical Tests:

Despite the mathematical conclusions for both the two-sample t-test and the two-sample t-interval, a formal conclusion regarding the initial hypotheses cannot be drawn because of the failure to meet the 10% condition for the early group. This is because the failure to meet the 10%

condition leads to the *possibility* that observations *might* be correlated with each other, distorting the standard error calculation. In other words, the computed standard error for the early group may not accurately reflect the true sampling variability, meaning the t-statistic, p-value, and interval become unreliable. Therefore, no conclusion can be drawn regarding whether the true mean ROI for early traders is different from that of late traders, or, if a difference does exist, how large it is.

Limitations & Potential Improvements:

The most obvious improvement is to use proportions instead of counts when sampling. Even though there were 5,080 wallets in the population, there were only 502 classified as early and 4,578 classified as late before sampling from those two groups. Randomly selecting 250 wallets from each group represents 5.46% of the late group, but a staggering 49.80% of the early group. This ultimately made the 10% condition fail for the early group. If proportions were used instead, then the results of this study could be different (and conclusions from the tests could be drawn). An alternative solution (using counts) would have been to sample only 50 wallets from each group, as 50 is $< 10\%$ of the 502 wallets in the early group.

One limitation was discovering a feature of Polymarket too late in the data collection process to account for it within the ROI calculation. Wallets can "split" and "merge" shares. Splitting allows traders to convert \$1 into a full set of outcome shares (one share of "Yes" and one share of "No"), while merging does the opposite (wallets can combine one "Yes" share and one "No" share to withdraw \$1). These actions are untraceable as they occur outside of the standard order book trades (the data I was using to reconstruct each wallet's trade history in the specific market), so they do not have traditional "Buy" or "Sell" prices. The implications of this

are obvious—calculated ROIs could become either much higher or lower than they truly were.

Fortunately, I was able to code something to see *which* wallets had performed a split, and *which* wallets performed a merge. Out of the 500 wallets in the sample, only 15 performed at least one merge, and 0 performed any splits. Future studies should look more in depth for alternative data sources to account for merges/splits so they can calculate a more accurate ROI for each wallet.

Another limitation was handpicking a timestamp to split the two groups. Moving the timestamp would change which wallets were within each population to be sampled from, and therefore can produce different results. Future studies should look into alternative (ideally quantitative) methods to find a point to "split" the market. Trading volume (e.g. if 200,000 shares were traded throughout a market's lifetime, then split on the timestamp of when the 100,000th share was traded) was considered, but ultimately not used because it is extremely susceptible to "whales" (wallets purchasing a large quantity of shares) to move the timestamp. The goal of a splitting timestamp is to find the point *right before* major news is released to the public (and therefore corrects prices), and the initial idea behind volume (before realizing the issue with whales) was that increased trading activity would be indicative of major news releases. Perhaps some metric to capture the lowest average time between trades is feasible (which indicates either rapid buying or panic selling). Future studies should look into either this method or alternative methods to finding a better "splitting" point.